

7 Neurofibromatosis type 1 (NF1) is a genetically inherited autosomal condition in which tumours grow in the nervous tissue.

- (a) If both parents of a child do **not** have NF1, then the chance that the child will have inherited NF1 is 0%.

In a family where one parent has NF1 and the other parent is unaffected, each child has a 50% chance of having NF1.

Using appropriate symbols, complete Figure 7.1 to show how there is a 50% chance of each child in this family having the condition.

symbols:		
parental phenotypes:	parent with NF1	unaffected parent
parental genotypes:		
gametes:		
offspring genotypes:		
offspring phenotypes:		

Figure 7.1

[4]

- (b) Genetic screening during pregnancy can help identify the risk of a fetus having NF1. Screening tests can include blood tests, ultrasounds and other procedures.

Discuss the social and ethical considerations of using genetic screening for NF1.

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- (c) One symptom of NF1, which can develop around the age of 20 years, is hearing loss caused by tumours of the auditory nerve. The auditory nerve contains sensory neurones that transmit impulses to the brain.

Suggest how a tumour on the auditory nerve could prevent the transmission of nerve impulses to the brain.

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